

The Rigid Grading: Layer Inequivalence, the Cross-Split Marriage, and the Algebra's Pre-Order on Mass Channels

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Abstract. The mass paper's Klein anatomy left the three splitting condensates $C \in \{4, 8, 12\}$ standing side by side, and the selection-principle analysis asked what could distinguish them. This note answers with three theorems, and the answer reshapes the summit: the second Klein grading — the layer grading, noted in the generation paper as "structure awaiting employment" — is *rigid*, and the algebra pre-orders the mass channels before the state chooses anything. **Theorem 1 (the split-relative anatomy):** the three splitters are the Cayley–Dickson tower's own doubling units — e_4 (the $\mathbb{H} \rightarrow \mathbb{O}$ step), e_8 (the $\mathbb{O} \rightarrow \mathbb{S}$ step), and their product e_{12} — and their marriages relate differently to the split $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}e_8$: the $C=4$ condensate pairs directions *within* each side, while $C=8$ and $C=12$ pair every direction with a partner *across* the divide (verified pair by pair). The complement enters mass exactly here: **the observable side is massed by its wedding to $\mathbb{O}e_8$.**

Theorem 2 (the trichotomy): the three channel algebras $\mathbb{H} + \mathbb{H}e_4 = \mathbb{O}$, $\mathbb{H} + \mathbb{H}e_8$, $\mathbb{H} + \mathbb{H}e_{12}$ — three eight-dimensional subalgebras sharing the same quaternions, the perfect mirror of the three generations sharing the core — are *not* three copies of one thing: the first two satisfy the composition law at 10^{-14} and are genuine octonion algebras, while $\mathbb{H} + \mathbb{H}e_{12}$ violates composition at order 34 — not an octonion, not a division algebra, a different kind of object. **Theorem 3 (rigidity):** the monomial automorphism search that found the exact generation $\mathbb{Z}_3$ finds, for the layers, *nothing*: no layer three-cycle, no layer swap, and no combined generation-layer cycle lifts to an automorphism (0 successes across all sign assignments). The generations are relative; the layers are absolute. Consequence for the selection principle: hierarchy's channels are algebraically graded — one intra-side octonionic channel, one cross-split octonionic channel, one cross-split non-composition channel — so the structure question of the summit is not pure spontaneous choice: the algebra supplies a pre-order, and the state chooses magnitudes within it.

1. The question

The Klein anatomy of the mass paper classified condensates: the unit gives universal mass, the imaginary core $C \in \{4, 8, 12\}$ splits within generations, the charged grades mix them. The summit analysis then asked the selection question, and one of its parts — *which* directions condense — turned on whether the three splitters are equivalent. The generation precedent suggested symmetry: the three generations are cycled by an exact automorphism. This note tests the layer analogue and finds the opposite.

2. Theorem 1: the splitters and the split

The splitting directions are not arbitrary core elements: e_4 is the doubling unit of the $\mathbb{H} \rightarrow \mathbb{O}$ step, e_8 of the $\mathbb{O} \rightarrow \mathbb{S}$ step, and $e_{12} = \pm e_4 e_8$; the neutral core is the algebra *generated by the tower's own doubling units* — the construction's history, sitting inside $\mathbb{S}$ as the shared quaternionic channel. Their marriages sort by the split (verified over all pairs):

$C = 4$: $(A, A \oplus 4)$ always same-side; $C = 8, 12$: $(A, A \oplus C)$ always cross-split.

The e_8 -condensate is literally the map that weds each observable-side direction to its complement image; the e_{12} -condensate does the same with a layer twist. The intuition that $\mathbb{O}_{e_8}$ "comes into play" for mass is thereby a theorem: *the complement enters mass through the cross-split marriages, and only there among the splitters.*

3. Theorem 2: the trichotomy of channel algebras

Each splitter generates, with the quaternions, an eight-dimensional subalgebra: $\mathbb{H} + \mathbb{H}e_4 = \mathbb{O}$, $\mathbb{H} + \mathbb{H}e_8$, and $\mathbb{H} + \mathbb{H}e_{12}$ — three subalgebras sharing $\mathbb{H}$, mirroring the three generation octonions sharing the core. The mirror breaks at the first test. Composition $N(xy) = N(x)N(y)$ on random elements: $\mathbb{O}$ passes at 2.8×10^{-14} ; $\mathbb{H} + \mathbb{H}e_8$ passes at 7.1×10^{-15} — an honest octonion algebra, the quaternions doubled by the sedenion unit; but $\mathbb{H} + \mathbb{H}e_{12}$ *fails at order 34*, with left-multiplication singular values dipping toward degeneracy: not a composition algebra, not a division algebra, a structurally different object. The three mass channels are three different kinds of algebra. ■

4. Theorem 3: the layers admit no symmetry

The same complete monomial search that exhibited the generation $\mathbb{Z}_3$ — index maps linear on the bits, now fixing the generation bits and permuting the layer bits, against all 2^{15} sign assignments — finds *nothing*: zero automorphisms among all layer three-cycles, zero among all layer swaps, and zero among combined generation-and-layer cycles. (For the pair $\mathbb{O} \leftrightarrow \mathbb{H} + \mathbb{H}e_8$ — both octonions — non-monomial maps are not excluded by this search; for any map moving $\mathbb{H} + \mathbb{H}e_{12}$ onto an octonionic channel the obstruction of Theorem 2 is invariant-level and absolute.) The two Klein gradings of $\mathbb{S}$ thus have opposite characters: **the generation grading carries an exact cyclic symmetry; the layer grading carries none.** Generations are relative; layers are absolute. ■

5. What the summit gains

The structure question of the selection principle — which condensates the vacuum populates — is refashioned. The generation side remains as the grading paper left it: exact democracy, hierarchy only by spontaneous $\mathbb{Z}_3$ breaking in the state. But the layer side, where the *splittings* live, is pre-ordered by the algebra itself: an intra-side octonionic channel (e_4), a cross-split octonionic channel (e_8), and a cross-split non-composition channel (e_{12}) — three inequivalent options that no symmetry can rotate into one another. Any dynamics built through the algebra's own vertices inherits this grading: the composition failure of the third channel is exactly the kind of structural difference that higher (associator-sensitive) vertices feel, so the inequivalence has a route into the effective potential — the mechanism by which the algebra's pre-order becomes a dynamical preference is identified, and belongs to the graded gap system of the road map's next step. The division of labor sharpens once more: the state still chooses the *numbers*; but the menu it chooses from was written by the algebra, and the menu is not symmetric.

6. Scope

Five items. (i) The composition and automorphism results are exact finite computations; the non-monomial gap is confined to the $\mathbb{O} \leftrightarrow \mathbb{H} + \mathbb{H}e_8$ pair and flagged. (ii) The dynamical realization of the pre-order — channel-dependent gap equations through associator-sensitive vertices — is the named next construction, not assumed here. (iii) The relation of the two octonionic channels' distinct embeddings to the classification results of the sedenion-split paper is noted for consolidation. (iv) Zero divisors of the third channel algebra are indicated by its singular-value behavior and not yet exhibited explicitly. (v) The summit's remaining parts — the vacuum-branch energy selection and the seed scale — stand as mapped in the road ahead.

References

- Companion papers: *Mass as Marriage: The Chirality Pairing, the Unit Condensate, and the Klein Anatomy of Mass Matrices*; *The Generation Grading: The Klein Selection Rule, Democratic Gauge Universality, and the Cyclic Generation Symmetry*; *The Sedenion Split $S = \mathbb{O} \oplus \mathbb{O} \cdot e_8$: Necessity, Classification, and Discrete Relativity*; *The Graded Net: Klein-Charged Currents, the Jordan Diagonality of Observables, and the Selection Rule as Scattering*.

Created in collaboration with Claude (Anthropic). All conclusions and any errors remain the author's responsibility.